

Week 5 Case Study: ADCS

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Part 1

Certification Practice Statement for your organization as part of the planning stage for implementing a PKI.

Identity of the CA

Security practices for maintaining CA integrity

Types of certificates issued

The policies for issuing, revoking, recovering, and renewing certificates

The cryptographic algorithms that are used and why

The certificate lifetimes and the reasoning behind them

The renewal policy of the CA's certificate

Part 2

What is a certificate?

A certificate is like an identification card for people, computers or websites on the internet. Certificates prove who someone or something is such that you can trust that you are working with the right person or thing. Certificate Authority is a trusted group that checks the details of a certificate and verifies them.

What is the difference between plaintext and ciphertext?

Plaintext is your message in a normal, readable form. An example of plaintext is a note saying “Hello World” that anyone can read right away. Ciphertext is that same message after it has been altered into secret code. An example of ciphertext is “Abbc67fzTN8tn”, and only people with the correct instructions can unscramble it back to its normal, readable form. Ciphertext hides the meaning of the message until it reaches someone trusted while plaintext can be read by anyone.

What is a key used for?

A key is used as a tool for scrambling a message to keep it safe from others, then used again to unscramble the message later so the correct person can read it. The message stays hidden in an unreadable and meaningless format without the matching key. This makes it so only trusted people can access the important information within that message.

What is the difference between a secret, private, and public key?

Secret keys are hidden codes shared between trusted people to access a message.

Private keys are yours and pairs with a matching public key to unlock messages sent to you or to prove that you are the real sender.

Public keys are shared openly so that anyone can use it to scramble messages for you, but only your private key can unscramble them.

Why does an organization need a certificate authority?

An organization needs a certificate authority to check if people are who they claim and to verify real identities, so teammates and customers know it is safe to connect. Without the certificate authority, anyone can impersonate people at a company and trick people into sharing secrets or fake messages.

What is the difference between an Enterprise Certificate Authority and a Stand-Alone Certificate Authority?

Enterprise Certificate Authorities work within a company network to hand out identities to trusted users and devices automatically without needing to ask a staff each time.

Standalone Certificate Authorities run independently and require a manager to review and approve all requests before issuing identities.

What is the difference between a one-level, two-level, and three-level CA Hierarchy?

A one-level hierarchy uses a single top authority that hands out identities to users and devices. There are no middle layers involved in this hierarchy.

A two-level hierarchy adds an intermediate authority below the top authority. The top authority allows the intermediate authority to hand out identification while the root is kept offline and safe.

A three-level hierarchy has a top authority, an intermediate authority, and a lower authority that gives identification which is suitable for big organizations that require security.

What hierarchy will your organisation use as part of your PKI design and why?

Our medical logistics company that delivers supplies by drone uses a two-level certificate authority hierarchy to protect the top authority while the middle level authority hands out identities. This suits our business because our top-level authority contains patient data and drone controls. At the same time, the intermediate authority will handle needs for devices and staff logins without risking the top authority.

We do not want to use a one-level certificate authority hierarchy in our organization because we would like to avoid the risk of being attacked at a single point of top-level authority.

Will your organisation use an Enterprise or Stand-Alone CA, and why?

Our organization will use an Enterprise Certificate Authority as our organization requires fast and automatic handling of giving out identities for users and devices without manual intervention that would slow down our operations. An Enterprise Certificate Authority scales well with growth because it also automatically handles more users and devices due to the way the technology is designed.

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